ASSIGNMENT 2

INSTRUCTIONS: STATE THE MOST SUITABLE DISTRIBUTION TO SOLVE THE FOLLOWING PROBLEMS WITH A SIMPLE REASON OF CHOICE OF DISTRIBUTION ATTACHED. SOLVE FOR THE ANSWER TOO. INTERPRET YOUR FINAL ANSWER FOR EACH QUESTION.

1. **Clinical Trial Success Rate**
   Problem: In a clinical trial of a new drug involving 20 patients, what is the probability that at least 15 patients will show a positive response to the treatment?

2. **ER Arrival Rate during Flu Season**

   Problem: An emergency room typically sees an average of 8 patients with flu-like symptoms per hour during the flu season. What is the probability that exactly 12 patients with flu-like symptoms will arrive in a given hour?

3. **Number of Attempts to Find a Matching Bone Marrow Donor**

   Problem: The probability of finding a matching bone marrow donor for a patient is 0.3. What is the probability that the first match is found on the 5th attempt?

4. **Modeling the Number of Drug Reactions**

   Problem: A pharmaceutical company expects an average of 2 adverse drug reactions per 1000 patients. If they administer the drug to 500 patients, what's the probability of observing more than 3 adverse reactions?

5. **Modeling Number of Patients Until 10 Recoveries**

   Problem: Suppose the probability of a patient recovering from a certain disease using a new treatment is 0.6. What is the probability that it will take exactly 15 patients to achieve 10 recoveries?

6. **Incidence of a Rare Genetic Mutation**

   Problem: A rare genetic mutation is present in 1 out of every 500 individuals. In a sample of 2000 individuals, what is the probability that exactly 3 people have the mutation?

7. **Modeling the Number of People Tested Until a Positive Case**

   Problem: During a disease outbreak, the probability of any randomly selected individual testing positive for the disease is 0.05. What is the probability that the first positive case is found after testing exactly 20 people?

8. **Modeling Number of Weeks to Observe 5 Cases of a Rare Disease**

   Problem: A rare disease occurs at an average rate of 1 case per week in a city. What is the probability that it will take at least 8 weeks to observe the 5th case?

STATE THE DISTRIBUTION TO BE USED AND THE REASON FOR THAT CHOICE. REASONS SHOULD BE BRIEF AS POSSIBLE.

9. Antibiotic Treatment Success
   Context: Phase III clinical trial with binary outcome

10. Hospital Infection Outbreaks
    Context: Monthly MRSA cases in ICU